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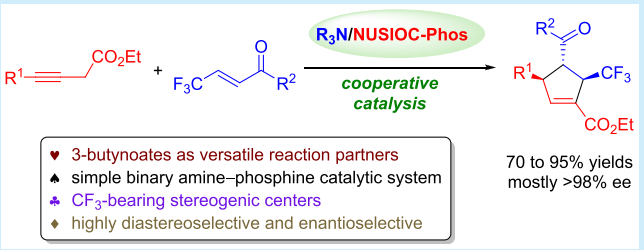
Highly Enantioselective [3 + 2] Annulation of 3-Butynoates with β-Trifluoromethyl Enones Promoted by an Amine–Phosphine Binary Catalytic System

Huanzhen Ni, Yee Lin Wong, Mingyue Wu, Zhaobin Han, Kuiling Ding,* and Yixin Lu*

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ABSTRACT: We report a highly enantioselective [3 + 2] annulation between 3-butynoates and β-trifluoromethyl enones, furnishing trifluoromethylated cyclopentenes with three contiguous stereogenic centers in good yields, high diastereoselectivities, and excellent enantioselectivities. A unique catalytic system consisting of a simple amine and a chiral phosphine was devised, and the synergistic play of Lewis basic amine and phosphine was crucial for alkyne isomerization and subsequent cyclization. The protocol disclosed herein allows facile activation of 3-butynoates in phosphine-mediated asymmetric transformations.



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Over the past decades, enantioselective nucleophilic phosphine catalysis has emerged as a powerful approach to access a wide range of optically enriched cyclic structural motifs.[1] Ever since Lu's pioneering work on phosphine-catalyzed [3 + 2] annulation between electron-deficient allenes and activated alkenes,[2a] the field of phosphine catalysis thrived and a broad range of cyclization methods have been developed over the years, including [3 + 2],[2] [4 + 1],[3] [4 + 2],[4] and [4 + 4][5] annulation reactions, among others.[6] Electron-deficient allenes are undoubtedly the most common substrates among all the suitable reaction partners in phosphine catalysis. In stark contrast, alkynes are much less employed, likely due to the difficulty associated with their activations. From a practical viewpoint, alkynes are readily available; therefore, broad utilization of alkyne substrates in phosphine-mediated annulation reactions would be highly desirable. Among different alkynes, the best studied substrates are 2-butynoate-type alkynes. Although 2-butynoates were more often used in asymmetric γ-additions,[7] their utilizations in asymmetric annulations are rare.[8] Ynones are unique alkynes possessing an acidic α′-proton, which were utilized as a C_3 synthon in a handful of [3 + 2] annulations by the groups of Fu,[9a] Shi,[9b] Huang,[9c] and Guo.[9d] In comparison with 2-butynoates and ynones, 3-butynoates were even less utilized and this is somewhat surprising given the fact that this class of alkynes are synthetically readily accessible.[10] To the best of our knowledge, there was only one report to date by Loh and co-workers in which they described the utilization of 3-butynoate in (*R*,*R*)-DIPAMP-catalyzed enantioselective [3 + 2] annulation with chalcones.[11] Apparently, a practical and general enantioselective cyclization strategy that makes use of 3-butynoates as a viable reaction partner for the creation of important chiral molecular architectures would be highly desirable, we therefore set our goal to tackle this challenge.

We reasoned the key to the utilization of 3-butynoates in phosphine catalysis is how to effectively activate this type of alkynes. In Loh's elegant example, highly nucleophilic bisphosphine (*R*,*R*)-DIPAMP was utilized for isomerizing 3-butynoates to the corresponding allenes and subsequently promoted the annulation reaction (Figure 1). We noticed that, in the literature reports, nitrogen Lewis base-catalyzed isomerization of 3-butynoates to allenoates was well documented.[12] We therefore envisioned that a binary catalytic system containing both an amine and a phosphine may offer a simple and practical solution to the activation of 3-butynoates. The achiral amine catalyst and chiral phosphine catalyst are

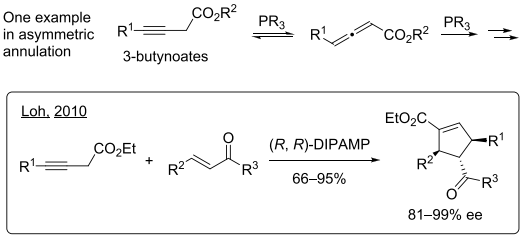


Figure 1. 3-Butynoates in asymmetric phosphine catalysis.

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anticipated to work in a synergistic manner; the amine facilitates the isomerization of alkyne substrate, and the resulting allenes will then be promoted by chiral phosphine for subsequent enantioselective transformations (Figure 2).

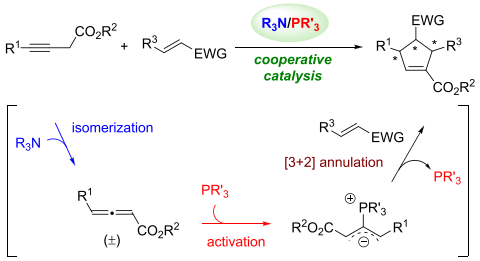


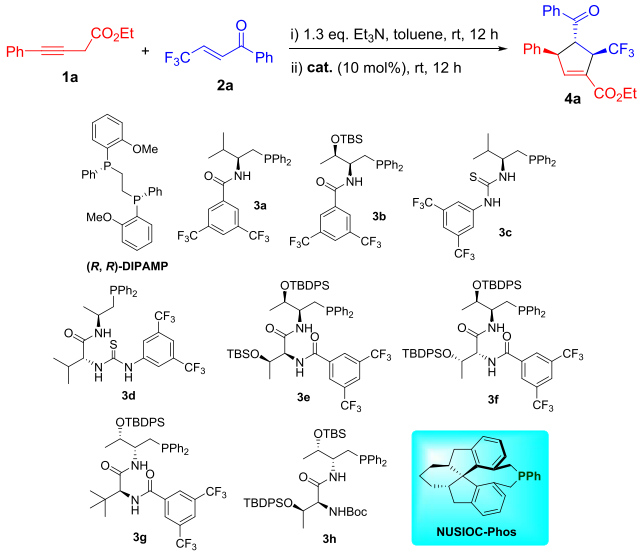
Figure 2. Our working hypothesis.

Moreover, the possibility of combining different amines and various phosphines adds in another attractive feature to this proposed strategy, a tunable system will enable different transformations of 3-butynoates, thus making them broadly applicable in phosphine-catalyzed asymmetric reactions.

Organofluorine compounds are molecules of great importance in pharmaceutical industry and agrochemistry, and their asymmetric synthesis is one of the most actively pursued areas in organic synthesis. Chiral cyclopentenes containing a trifluoromethylated stereogenic center are structural scaffolds that are widely present in many bioactive molecules,[13] and an efficient method for creating CF_3-containing stereogenic centers has always been well-sought.[14] Herein, we demonstrate that simple combination of achiral amine and chiral phosphines constitutes a powerful catalytic system for the activations of 3-butynoates, leading to the creation of optically enriched cyclopentenes bearing a CF_3-containing stereogenic center.

We started our investigations by examining the reaction between 3-alkynoate **1a** and enone **2a** in the presence of amine/phosphine binary catalytic systems (Table 1). We first tested the annulation reaction with (*R*,*R*)-DIPAMP,[11] and poor dr, moderate yield, and enantioselectivity were observed (entry 1). Subsequently, we treated 3-alkynoate **1a** with triethylamine for 12 h, after which enone **2a** and (*R*,*R*)-DIPAMP were introduced. With this protocol, the [3 + 2] annulation proceeded smoothly and cyclopentene **4a** with three contiguous stereogenic centers was obtained in good yield and ee (entry 2). We then turned to our amino acid-derived bifunctional phosphines. Both L-valine-derived catalyst **3a** and L-threonine-derived phosphine-amide **3b** were ineffective in asymmetric induction and so were phosphine-thioureas **3c** and **3d** (entries 3−6). Dipeptide-based phosphines **3e**−**3h** led to much improved enantioselectivities but still only modest at most (entries 7−10). When our recently developed NUSIOC-Phos[15] was evaluated, we were delighted to discover that the [3 + 2] annulation product **4a** was formed in high chemical yield, with good diastereoselectivity, and with 99% enantiomeric excess (entry 11). Next, we evaluated the necessity of having both amine and phosphine in the catalytic system. When the reaction was performed in the presence of only NUSIOC-Phos, a trace amount of product was detected, suggesting the important role that triethylamine had played in the reaction (entry 12). If the reaction was performed without NUSIOC-Phos, only isomerization to the corresponding γ-substituted allenoate was

Table 1. Initial Screening[a]



entry	phosphine	dr[b]	yield (%)[c]	ee (%)[d]
1[e]	(*R*,*R*)-DIPAMP	3:1	70	70
2	(*R*,*R*)-DIPAMP	>20:1	90	84
3	**3a**	7:1	82	12
4	**3b**	6:1	74	12
5	**3c**	>20:1	95	18
6	**3d**	>20:1	91	24
7	**3e**	1:2	85	77
8	**3f**	6:1	76	70
9	**3g**	8:1	81	−65
10	**3h**	9:1	84	30
11	NUSIOC-Phos	12:1	87	99
12[f]	NUSIOC-Phos	–	trace	–
13[g]		–	–	–
14[h]	NUSIOC-Phos	11:1:2	75	96

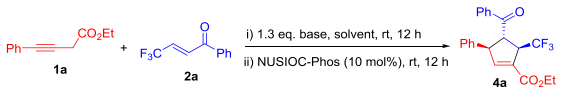
[a]Reactions were performed by treating **1a** (0.12 mmol) in toluene (1 mL) with Et_3N (0.13 mmol) and stirred for 12 h at room temperature, followed by the addition of **2a** (0.10 mmol) and the catalyst (10 mol %). [b]Determined by crude ^1H NMR analysis. [c]Isolated yield of the major diastereomer. [d]Determined by HPLC analysis on a chiral stationary phase. [e]The reaction was performed with **1a**, **2a**, and (*R*,*R*)-DIPAMP (10 mol %) in toluene. [f]The reaction was carried out in the presence of NUSIOC-Phos. [g]The reaction was performed in the presence of only Et_3N. [h]One-pot reaction was carried out.

observed (entry 13). Lastly, we carried out one-pot reaction and the results were less satisfactory. When triethylamine and phosphine were added in a one-step, one-pot fashion, although **4a** was obtained in good yield and excellent ee value, another diastereomer was formed during the reaction (entry 14).

The effects of different nitrogen bases were next evaluated (Table 2). Compared with triethylamine, *N*,*N*-diisopropylethylamine (DIPEA) displayed virtually the same catalytic activities, and high yield and an excellent ee value were attainable (entry 2). Surprisingly, the employment of 1,4-diazabicyclo[2.2.2]octane (DABCO) led to very poor yield of the desired product. In addition, another unidentified diastereomer was formed as the major product (entry 3). Inorganic base, such as Cs_2CO_3, could also be used; while diastereo- and enantio- selectivities remained excellent, the yield was only modest (entry 4). Subsequent solvent screening

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Table 2. Examination of Base and Solvent[a]



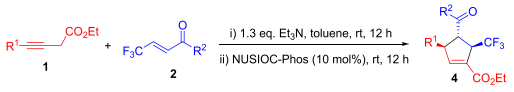
entry	base	solvent	dr[b]	yield (%)[c]	ee (%)[d]
1	Et_3N	toluene	12:1	87	99
2	DIPEA	toluene	12:1	85	98
3	DABCO	toluene	5:1:17	14	–
4	Cs_2CO_3	toluene	10:1	71	99
5	Et_3N	EtOAc	9:1	80	98
6	Et_3N	CH_2Cl_2	3:1	67	76
7	Et_3N	$CHCl_3$	4:1	72	98
8	Et_3N	dioxane	11:1	84	97

[a]Reactions were performed by treating **1a** (0.12 mmol) in solvent (1 mL) with the base specified (0.13 mmol) and stirred for 12 h at room temperature, followed by the addition of **2a** (0.10 mmol) and NUSIOC-Phos (10 mol %). [b]Determined by ^{1}H NMR analysis of the crude mixture. [c]Isolated yield of the major diastereomer. [d]Determined by HPLC analysis on a chiral stationary phase.

was performed, and toluene remained to be the most suitable solvent for this reaction (entries 5−8). Notably, the reaction also proceeded smoothly with 20 mol% Et_3N employed in the first step, leading to the formation of the annulation product with virtually the same efficiency.

With the optimized reaction conditions in hand, we proceeded to examine the general applicability of the method (Table 3). Trifluoromethylated enones **2** bearing different aryl substituents were excellent substrates, and highly functionalized cyclopentenes were obtained in good yields, with high diastereoselectivities and nearly perfect enantioselectivities (entries 1−12). Furthermore, different aryl-substituted 3-butynoates were also found to be suitable, and good yields, high diastereoselectivities, and extremely high enantioselectivities were attainable (entries 13−20). The absolute configurations of cyclopentenes **4** were assigned by analogy based on the X-ray crystallographic analysis of product **4s**.[17] The employment of alkyl-substituted 3-butynoate or β-trifluoromethyl enones in the reaction failed to yield the desired product.[16]

Table 3. Reaction Scope[a]



entry	R^1 (**1**)/R^2(**2**)	**4**	dr[b]	yield (%)[c]	ee (%)[d]
1	Ph (**1a**)/Ph (**2a**)	**4a**	12:1	87	99
2	Ph (**1a**)/4-ClC_6H_4 (**2b**)	**4b**	13:1	89	99
3	Ph (**1a**)/3-ClC_6H_4 (**2c**)	**4c**	13:1	90	99
4	Ph (**1a**)/2-ClC_6H_4 (**2d**)	**4d**	10:1	75	91
5	Ph (**1a**)/4-BrC_6H_4 (**2e**)	**4e**	16:1	80	98
6	Ph (**1a**)/4-FC_6H_4 (**2f**)	**4f**	15:1	92	99
7	Ph (**1a**)/4-CNC_6H_4 (**2g**)	**4g**	15:1	90	95
8	Ph (**1a**)/4-MeC_6H_4 (**2h**)	**4h**	14:1	91	99
9	Ph (**1a**)/4-$MeOC_6H_4$ (**2i**)	**4i**	8:1	81	99
10	Ph (**1a**)/3,4-$(MeO)_2C_6H_3$ (**2j**)	**4j**	13:1	72	99
11	Ph (**1a**)/2-naphthyl (**2k**)	**4k**	17:1	95	98
12	Ph (**1a**)/2-thienyl (**2l**)	**4l**	16:1	93	98
13	4-ClC_6H_4 (**1b**) /Ph (**2a**)	**4m**	9:1	78	99
14	3-ClC_6H_4 (**1c**) /Ph (**2a**)	**4n**	8:1	73	98
15	4-FC_6H_4 (**1d**) /Ph (**2a**)	**4o**	10:1	81	99
16	2-FC_6H_4 (**1e**) /Ph (**2a**)	**4p**	8:1	70	99
17	4-BrC_6H_4 (**1f**) /Ph (**2a**)	**4q**	8:1	71	98
18	4-EtC_6H_4 (**1g**) /Ph (**2a**)	**4r**	15:1	89	99
19	4-MeC_6H_4 (**1h**) /Ph (**2a**)	**4s**	16:1	88	99
20	4-$MeOC_6H_4$ (**1i**) /Ph (**2a**)	**4t**	12:1	82	99

[a]Reactions were performed by treating **1** (0.12 mmol) in toluene (1 mL) with Et_3N (0.13 mmol) and stirred for 12 h at room temperature, followed by the addition of **2** (0.10 mmol) and NUSIOC-Phos (10 mol %). [b]Determined by ^{1}H NMR analysis of the crude mixture. [c]Isolated yield of the major diastereomer. [d]Determined by HPLC analysis on a chiral stationary phase.

The isomerization process of 3-butynoate **1a** facilitated by triethylamine was studied by ^{1}H NMR analysis. The freshly prepared 3-butynoate **1a** contains 7.6% of allenoate **1a′**. With the addition of triethylamine, the percentage of allenoate in the system increased rapidly within the first few hours and then reached equilibrium (83%) at around 12 h (Supporting Information). When NUSIOC-Phos and enone **2** were introduced to the system, the [3 + 2] annulation took place, which further shifted the equilibrium between 3-butynoate **1a** and allenoate **1a′** toward creation of more **1a′**, until **1a** was fully consumed.

In conclusion, we introduced a novel concept of utilizing an amine−phosphine binary catalytic system for effective activation of 3-butynoate substrates in phosphine catalysis. In the presence of triethylamine and NUSIOC-Phos, the [3 + 2] annulation between 3-butynoates and trifluoromethylated enones proceeded smoothly, affording trifluoromethylated cyclopentenes with three contiguous stereogenic centers in good yields, high diastereoselectivities, and nearly perfect enantioselectivities. Notably, the method disclosed herein allows for general and potentially broad application of 3-butynoates in phosphine-mediated asymmetric annulation reactions.[18] Our investigations in this direction are ongoing.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.0c00681.

Experimental details, analytical data and HPLC chromatograms of products, and copies of NMR spectra (PDF)

Accession Codes

CCDC 1902409 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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(16) When *n*-Pr-substituted 3-butynoate was employed to react with enone **2a**, only a trace amount of product was observed. When phenethyl-substituted *β*-trifluoromethyl enone was reacted with 3-butynoate **1a**, no desired annulation product was observed.

(17) CCDC 1902409 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

(18) See the Supporting Information for our preliminary studies of phosphine-catalyzed [3 + 2] annulation between 3-butynoate and chalcones.

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